INCH-POUND

A-A-59440/5B
21 May 2012
SUPERSEDING
A-A-59440/5A
23 May 2005

COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

COCK-VALVE, DRAIN, GROUND KEY, BIB NOZZLE 150 PSI, STEAM PRESSURE, BRASS, NPTF-1 THREADS

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

The complete requirements for procuring the cock-valve, drain, ground key, bib nozzle, 150 psi, steam pressure, brass, NPTF-1 threads described herein shall consist of this document and the issue in effect of A-A-59440.

CLASSIFICATION/PART OR IDENTIFICATION NUMBER (PIN). This commercial item description (CID) specification sheet uses a classification system which is included in the PIN as shown in the following example (see NOTES).

AA59440	/05	-	001
CID number	Applicable CID spec sheet (two digits)		Dash number (see figure 1)

SALIENT CHARACTERISTICS.

Interface and physical dimensions. Cock-valves, drain, ground key, bib nozzle, 150 psi, steam pressure, brass, NPTF-1 threads supplied to this CID specification sheet shall be as specified herein (see figure 1).

Sizes. Nominal sizes of 1/8 inch through 1 inch with NPTF-1 threads.

Intended applications. This item is for use in steam systems where pressures do not exceed 150 psig (10.3 bar) at 351°F (177°C) and the liquid is compatible with brass.

Permissible variations in dimensions. Figure 1 lists the permissible variation in dimensions for each size. The external dimensions of the cock-valve should not limit procurement unless the end item applications restrict these dimensions.

Valve bodies. Valve bodies shall be provided with hexagonal wrenching surfaces.

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Threads. Threads are dryseal NPTF-1 in accordance with A-A-59440.

Material.

Preferred valve material. Preferred material is brass in accordance with ASTM B61.

Alternate brass. If other than the preferred brass is used the alternate brass shall be in accordance with the following commercial standards:

ASTM B16/B16M	ASTM B121/B121M	ASTM B369	SAE AMS4845
ASTM B22	ASTM B124/B124M	ASTM B505/B505M	SAE AMS4855
ASTM B30	ASTM B148	ASTM B584	SAE AMS4860
ASTM B36/B36M	ASTM B176	ASTM B763	SAE AMS4862
ASTM B62	ASTM B271	ASTM B770	SAE AMS4890

Protective coating.

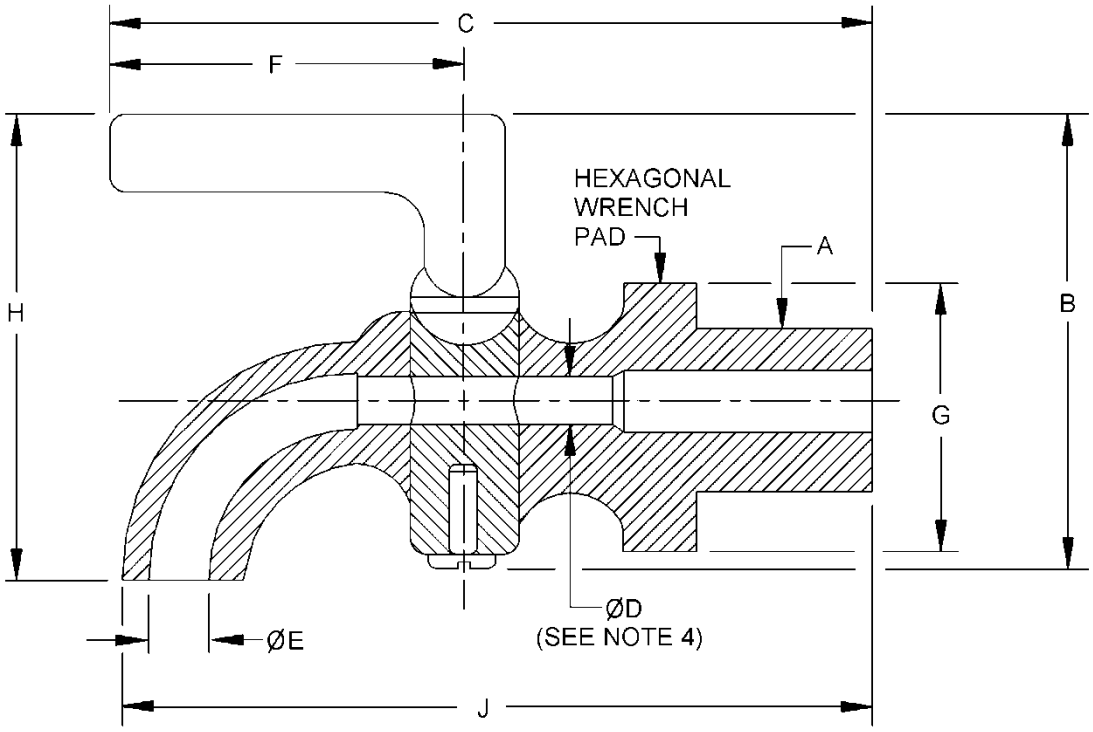
Brass: none.

Caution:
Brass shall not have any lead contents.

Test pressures.

a. Shell: 300 psig (20.7 bar).
b. Seat: 150 psig (10.3 bar).

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Dash number	A pipe thread NPTF-1	B max (mm)	C max (mm)	D diameter min (mm)	E diameter min (mm)	F max (mm)	G across flats ±.015 (0.38 mm)	H max (mm)	J max (mm)
021	1/8-27	1.81 (46.0)	2.75 (70.0)	.14 (3.6)	.19 (4.8)	1.38 (35.1)	.50 or .69 (12.7 to 17.5)	1.81 (46.0)	2.50 (63.5)
022	1/4-18	1.81 (46.0)	2.75 (70.0)	.16 (3.6)	.22 (5.6)	1.38 (35.1)	.59 or .69 (15.0 or 17.5)	1.81 (46.0)	2.50 (63.5)
023	3/8-18	2.94 (74.7)	4.09 (103.9)	.25 (6.4)	.28 (7.1)	2.25 (57.2)	.78 or .86 (19.8 or 21.8)	2.91 (73.9)	3.44 (87.4)
024	1/2-14	3.38 (85.9)	4.39 (111.5)	.31 (7.9)	.34 (8.6)	2.28 (57.9)	.97 or 1.00 (24.6 or 25.4)	3.53 (89.7)	3.61 (91.7)
025	3/4-14	3.80 (96.5)	4.81 (122.2)	.62 (15.8)	.56 (14.2)	2.50 (63.5)	1.06 or 1.25 (26.9 or 31.8)	3.94 (100.1)	4.50 (114.3)
026	1-11-1/2	4.56 (116.0)	6.31 (160.3)	.97 (24.6)	.94 (23.9)	3.44 (87.4)	1.25 or 1.50 (31.8 or 38.1)	4.75 (120.7)	5.62 (142.7)

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerances are ±.010 inch (0.25 mm).
4. Trapezoidal or oval port having same cross sectional area is acceptable.

FIGURE 1. Cock-valve, drain, ground key.

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NOTES.

PIN. The PIN should be used for Government purposes to buy commercial products to this CID specification sheet. See the classification section for PIN format example.

Source of documents.

COMMERCIAL ITEM DESCRIPTION

A-A-59440	-	Cock-Valves and Valves, General Requirements for

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

Other Publications

ASTM INTERNATIONAL

ASTM B16/B16M	-	Free-Cutting Brass Rod, Bar and Shapes for use in Screw Machines
ASTM B22	-	Bronze Castings for Bridges and Turntables
ASTM B30	-	Copper Alloys in Ingot Form
ASTM B36/B36M	-	Brass Plate, Sheet, Strip, and Rolled Bar
ASTM B61	-	Steam or Valve Bronze Castings
ASTM B62	-	Composition Bronze or Ounce Metal Castings
ASTM B121/B121M	-	Leaded Brass Plate, Sheet, Strip, and Rolled Bar
ASTM B124/B124M	-	Copper and Copper Alloy Forging Rod, Bar, and Shapes
ASTM B148	-	Aluminum-Bronze Sand Castings
ASTM B176	-	Copper-Alloy Die Castings
ASTM B271	-	Copper-Base Alloy Centrifugal Castings
ASTM B369	-	Copper-Nickel Alloy Castings
ASTM B505/B505M	-	Copper Alloy Continuous Castings
ASTM B584	-	Copper Alloy Sand Castings for General Applications
ASTM B763	-	Copper Alloy Sand Castings for Valve Applications
ASTM B770	-	Copper-Beryllium Alloy Sand Castings for General Applications

(Copies of these documents are available from www.astm.org or ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken, Pennsylvania, USA 19428-2959.)

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SAE INTERNATIONAL

SAE AMS4845	-	Tin Bronze Castings, Sand and Centrifugal 87.5CU - 10SN - 2ZN as Cast
SAE AMS4855	-	Leaded Red Brass, Sand and Centrifugal Castings 85CU - 5.0SN - 5.0PB - 5.0ZN as Cast
SAE AMS4860	-	Manganese Bronze, Sand and Centrifugal Castings 58CU - 39ZN - 1.2FE - 1.0AL - 0.80MN as Cast
SAE AMS4862	-	Manganese Bronze, Sand and Centrifugal Castings 63CU - 24ZN - 6.2AL - 3.8MN - 3.0FE High Strength, as Cast
SAE AMS4890	-	Copper-Beryllium Alloy Castings 97CU - 2.1BE - 0.52CO - 0.28SI Solution Heat Treated (TBOO)

(Copies of these documents are available on line at www.sae.org from SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001, and Tel: 877-606-7323 [inside USA and Canada] or 724-776-4970 [outside USA], email at CustomerService@sae.org.)

Commercial products. As part of the market analysis and research effort, this CID specification sheet was coordinated with the following manufacturers of commercial products. At the time of CID specification sheet preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID specification sheet. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

MFR's CAGE	MFR's name and address
01276	Eaton Aeroquip Inc. 14615 Lone Oak Rd. Eden Prairie, MN 55344 952-294-2032/888-258-0222 jeffenoe@eaton.com
93061	Parker-Hannifin Corp Brass Products Division 300 Parker Drive Otsego, MI 49708-1472 260-636-2104 dwsanders@Parker.com
20969	Essex Brass Corp 23500 Pinewood Warren, MI 48091-3122 586-757-8200 bpushman@essexbrass.com

Military standard (MS) supersession data. This CID replaces the following MS as shown in table I.

TABLE I. MS supersession data.

CID PIN	Superseded MS
AA59440/5-021	MS35787-1
AA59440/5-022	MS35787-2
AA59440/5-023	MS35787-3
AA59440/5-024	MS35787-4
AA59440/5-025	MS35787-5
AA59440/5-026	MS35787-6

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Part number P/N supersession data. These CID specification sheet PINs supersede the following MFR's P/N's as shown in table II. This information is being provided to assist in reducing proliferation in the Government inventory system.

TABLE II. P/N supersession data.

Dash number (see figure 1) AA59440/5	MFR's CAGE	MFR's P/N 1/
021	20969	SC29
022	20969	SC30
023	20969	SC31
024	20969	SC32
025	20969	N/A
026	20969	N/A

1/ The manufacturer's P/N shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see marking paragraph of A-A-59440.

National stock number (NSN). The following is a list of NSN's assigned which correspond to this CID specification sheet. The list in table III is for information only and may not be indicative of all possible NSN's associated with the CID specification sheet. For up to date information on assigned NSN's, please contact the aforementioned DLA Land and Maritime office listed in A-A-59440.

TABLE III. NSN's.

Dash number (see figure 1) AA59440/5	NSN
021	4820-00-272-3332
022	4820-01-193-0255
025	4820-00-272-3336

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

MILITARY INTERESTS:

Custodians:
Army - AR
Navy - SH
Air Force - 99
DLA - CC

Review activities:
Army - GL, MI
Navy – AS, MC, SA

CIVIL AGENCY AND COORDINATING ACTIVITIES:

GSA - FSS

Preparing activity:

DLA - CC

(Project 4820-2011-035)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organization and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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